HW 4.3

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NOTE: Coding throughout the assignment is done in the style of Tidyverse

1. Busiest Airport

Table

Passenger Traffic for Six Major Airports (2020–2025)

airport	2020	2021	2022	2023	2024
Beijing Daxing International Airport	NA	25,051,012	NA	NA	49,441,029
Frankfurt Airport	18,768,601	24,812,849	48,918,482	59,355,389	61,564,957
Hartsfield–Jackson Atlanta International Airport	42,918,685	75,704,760	93,699,630	104,653,451	108,067,766
Heathrow Airport	22,111,469	NA	61,614,508	79,183,364	83,884,572
Los Angeles International Airport	28,779,527	48,007,284	65,924,298	75,050,875	76,588,028
São Paulo/Guarulhos International Airport	20,359,547	24,164,330	34,466,946	41,307,919	NA

Plot/Graph

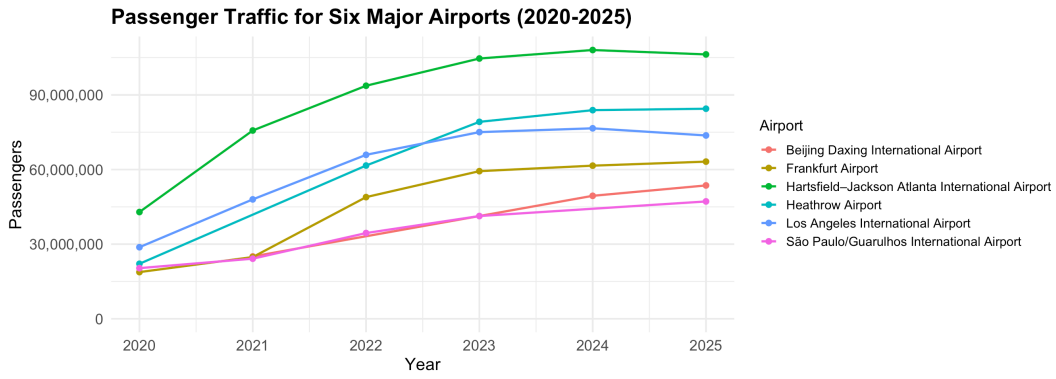

Figure 1: Passenger traffic trends across 6 of the 50 largest airports in the world (2020–2025)

Narrative Text

The table and plot both are examining the growth in passenger traffic for 6 of the 50 busiest airports in the world. Although we can't see the full view prior to 2020, all airports experienced a sharp decline or lower traffic levels, where travel became heavily restricted due to the Covid-19 Pandemic.

From the visualizations, we can analyze the situation as the starting point for each airport in 2020 was at its lowst in the 5-year span. Following 2020, all airports showed different levels of upward growth, indicating a recoery in passeneger traffic.

Hartsfield-Jackson Airport in Atlanta, Georgia handled the most passenger traffic of the 6 airports by far, which can be shown in the visualization, sitting at or exceeding 100 million passengers since 2023. Los Angeles International Airport and Heathrow Airport also showed strong signs of recovery due to their constant growth and steady increase.

There does exist some variation between airports. No airport declined in passenger travel after 2020, however some leveld off or saw less of a slower increase than others. São Paulo/Guarulhos International Airport in Brazil struggled to see the same growth, although it did increase.

Just a note, but some airports also have "NA" for passenger traffic. This was due to the fact that when scraping the data, some times the airports fell out of the top 50 rankings so they didn't have data on them.

Overall we can see that after 2020 all of the airports saw some kind of growth from 2020 to 2025 implying a positive recovery in global travel with consistent growth across all airports.

2. Monte Carlo Integration

Plot/Graph

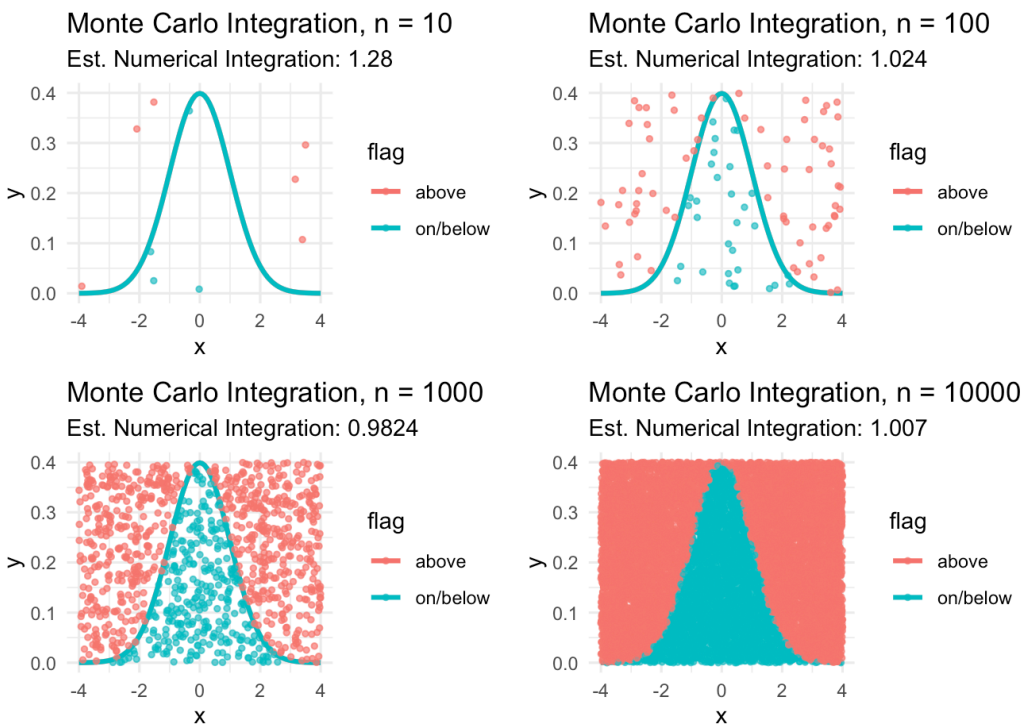

Figure 2: Monte Carlo Table with various different sample sizes to show it's growth in stability

Narrative Text

The Monte Carlo integration method estimates the area under a curve by randomly sampling points within a defined region and calculating what proportion of points fall below the function. In the four examples above, random points are generated uniformly across the domain and bounded vertically. Points are split up into two groups, on/below or above. The estimated proportion is then calculated based on the ratio.

We can see how the Monte Carlo Integration behaves when given a sample size of 10, 100, 1000, and 10000. As the sample size gets larger, we can see more stability and more clearly defined separation between the behaviors of the function. The density of points increases as the sample size increases, yet the proportion stays around the same in each example (approximately 1.00). The small multiple plots demonstrate this clearly, as the variabilitydecreases and the proportion of points under the

curve starts to stabalize and densify. The visualization supports the reasoning showing that a larger sample results in a more uniform and representative distribution of points across the area.

Overall, the simulation effective illustrates both the process of Monte Carlo integration, highlighting that sampling improves with accuracy.

3. Planning and Prompting Generative AI

Below is the plan that I developed and fed into ChatGPT to return the resulting R code to generate the given plot

Context + Goals

The ‘calcium.csv’ data set comes from a longitudinal study that followed 31 women over four years to measure the amound of calcium each of their ulnas in their dominant arms absorbed through photon absorptiometry.

The goals of the experiment are to: - Compare the calcium levels of each women’s dominant arm’s ulnas

- Compare trends between the null group and the diet/exercise group
- Determine whether there is a noticeable improvement relative to the null group.

Currently, the data set is not tidy as it is in a wide format, where each woman has multiple columns for different years and groups.

Steps to Tidy Data

Sub-Goal 1: Tidy the Data

Steps

1.) Identify columns coressponding to the null group or diet/exercise group and associate each entry with a time (Initial, year 1, year 2, year 3)

2.) Change the format of the data set from wide to long

3.) Create the following columns:

- (calcium) one column represent the calcium measurement
- (group) one column represent the treatment group, whether they were null or were dieting and exercising)
- (id) one column for the participant ID
- (year) time period of collection (Initial, year 1, year 2, year 3)

NOTE: There should be 136 rows

Sub-Goal 2: Clean and Validate the Data

Steps

1.) Make sure that the calcium values are accpeted into the csv file (i.e. numeric values in each cell)

2.) Make sure the labels are standardized

3.) Verify for duplicate or missing values

Visualization

1.) Use a line plot to visualize the changes over time (x-axis: year, y-axis: calcium level, color: Treatment group)

2.) We want to include the individual lines for each particiapnt at a lower opacity to show the variation and not clutter the data. An average trend line will then be place for each group to simplify the data as much as possible

3.) This graph will be able to the changes in calcium levels over time, the diferences in treatment groups, and whether there was a significant impact in dieting and exercising and calcium levels.

Resulting Plot/Graph

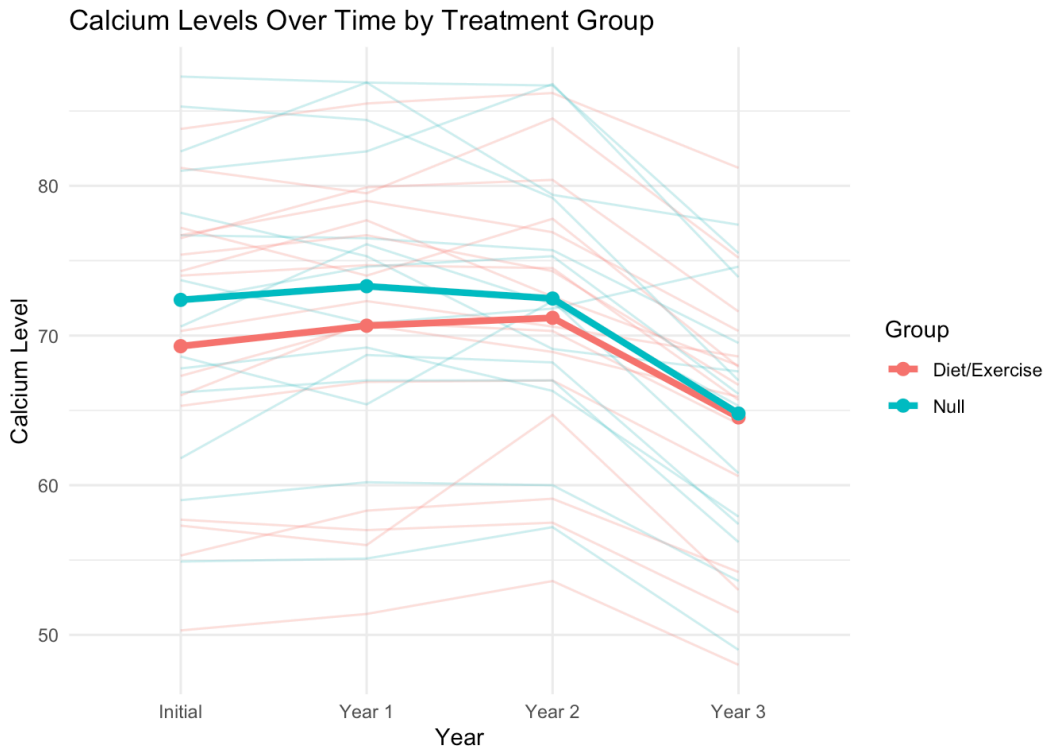

Figure 3: Calcim Levels Graph visualization output by AI follwoing plan

Visualization Output from Generic Prompt

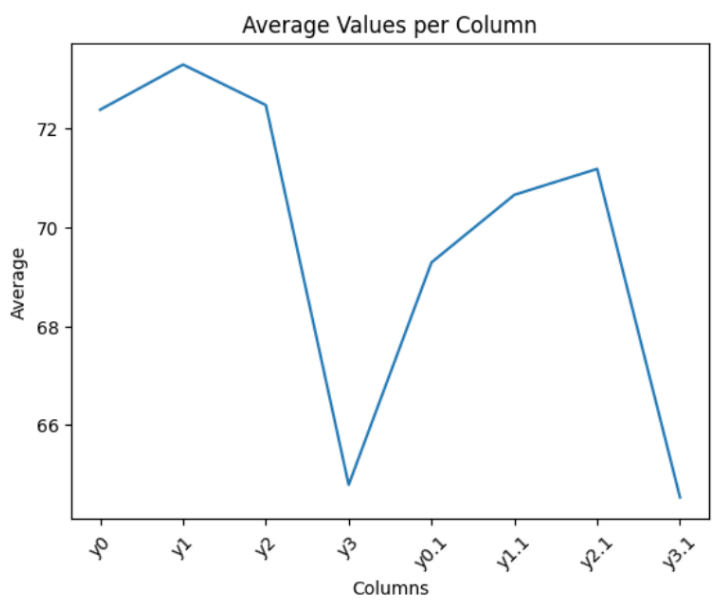

Figure 4. Calcium Levels over time generated from a generic AI prompt.

Narrative Text

This section compares the calcium dataset by looking at the developed plan and resulting visualization, a generic AI prompt and resulting output, and the interpretation of both.

The developed plan incorporates individual trajectories for each participant along with group-level summary lines, allowing for a more detailed understanding of both variability and overall trends. The faded lines lines show the calcium levels per person and the distinct color lines are the average of the members for each group, forming a trend line. From this visualization, it is clear that both groups follow a similar patern, with a slight increase in cacium for the first year and then a notable decline so by the end of the experiment the null and diet/exercise group are basically the same.

Looking at the generic prompt, the generic AI Prompt displays a horrible chart displaying a chart titled “Average Values per Column” where it doesn’t take into account the two groups, nothing is being compared, and overall, it’s hard to read and understand. It’s very clear to see that without setting up a plan, AI doesn’t know what it’s supposed to do, causing this very confusing output.

Although we could feed AI our data and context and ask it to analyze it for us, taking the time to plan and actually approach the problem can get us a lot more informed result that we can actually analyze. AI did a good job with what it was given, but it’s too surface level for the complexity of the problem.

Overall, the comparison demonstrates that while AI can quickly get us a visualization, it struggles in understanding the complexity, quality, and depth of the problem to output a proper result. Maybe in the future, AI will catch up, but at this point, human though process and intentional design is still the stronger method for developing and analyzing statistical work.

4. Self-Reflection

Although I had some experience working in R prior to taking STAT 184, I did end up learning a lot about the strengths and suits of the R language. For example, I am a Data Analyst for some sports teams on on campus so I’m mainly working with sports data, I found that scrapping data online is much easier in R rather than Python. I have tried web scraping in Python however I just can’t seem to get it right. R made it really easy to grab all the data and then from there, I also learned how to clean up the data and make it usable for analysis.

Another thing I am starting to learn how to use is GitHub. Although I have used GitHub before, I've only used it to save my progress. I have never made a branch or actually truly integrated it into my workflow, becaue most of my projects are solo; I treated GitHub as a 1-Dimensional cloud service. Now I am starting to learn how to use branches and other features that I can integrate into my workflow to make saving progress and sharing my work a lot easier.

5. Apendix

Usage 1: Creating the AI Graph in Question 3

Date of Usage

April 13, 2026

Prompts Used

"Please follow the instructions step by step to return the code to run the visualization. Please return only the code and no extra jargon"

```
## Context + Goals
The 'calcium.csv' data set comes from a longitudinal study that
followed 31 women over four years to measure the amound of
calcium each of their ulnas in their dominant arms absorbed
through photon absorptiometry.

The goals of the experiment are to:
- Compare the calcium levels of each women's dominant arm's
ulnas

- Compare trends between the null group and the diet/exercise
group

- Determine whether there is a noticable improvement relative
to the null group.

Currently, the data set is not dity as it is in a wide format,
where each woman has multiple columns for different years and
groups.

## Steps to Tidy Data
### Sub-Goal 1: Tidy the Data
```

```
#### Steps
1.) Identify columns coressponding to the null group or
diet/exercise group and associate each entry with a time
(Initial, year 1, year 2, year 3)

2.) Change the format of the data set from wide to long

3.) Create the following columns:

  - (calcium) one column represent the calcium measurement

  - (group) one column represent the treatment group, whether
they were null or were dieting and exercising)

  - (id) one column for the participant ID

  - (year) time period of collection (Initial, year 1, year 2,
year 3)

NOTE: There should be 136 rows

### Sub-Goal 2: Clean and Validate the Data
#### Steps
1.) Make sure that the calcium values are accpeted into the csv
file (i.e. numeric values in each cell)

2.) Make sure the labels are standardized

3.) Verify for duplicate or missing values

## Visualization
1.) Use a line plot to visualize the changes over time (x-axis:
year, y-axis: calcium level, color: Treatment group)

2.) We want to include the individual lines for each
particiapnt at a lower opacity to show the variation and not
clutter the data. An average trend line wil then be place for
each group to simplify the data as much as possible

3.) This graph will be able to the changes in calcium levels
over time, the diferences in treatment groups, and whether
there was a significant impact in dieting and exercising and
calcium levels.
```

Output

```
library(tidyverse)
library(knitr)
```

```
library(kableExtra)

calcium <- read_csv("calcium.csv", show_col_types = FALSE)

names(calcium) <- c(
  "null_initial", "null_year1", "null_year2", "null_year3",
  "diet_initial", "diet_year1", "diet_year2", "diet_year3"
)

calcium_tidy <- bind_rows(
  calcium |>
    select(starts_with("null")) |>
    mutate(id = row_number(), group = "Null") |>
    pivot_longer(
      cols = starts_with("null"),
      names_to = "year",
      values_to = "calcium"
    ),

  calcium |>
    select(starts_with("diet")) |>
    mutate(id = row_number(), group = "Diet/Exercise") |>
    pivot_longer(
      cols = starts_with("diet"),
      names_to = "year",
      values_to = "calcium"
    )
) |>
  mutate(
    year = case_when(
      str_detect(year, "initial") ~ "Initial",
      str_detect(year, "year1") ~ "Year 1",
      str_detect(year, "year2") ~ "Year 2",
      str_detect(year, "year3") ~ "Year 3"
    ),
    year = factor(year, levels = c("Initial", "Year 1", "Year
2", "Year 3")),
    calcium = as.numeric(calcium)
  ) |>
  drop_na(calcium)

calcium_tidy |>
  head(20) |>
  kable(caption = "Table 1. Tidy Calcium Dataset (First 20
Rows)") |>
  kable_styling(full_width = FALSE)

ggplot(calcium_tidy, aes(x = year, y = calcium, color = group))
```

```
+
  geom_line(aes(group = interaction(group, id)), alpha = 0.25,
na.rm = TRUE) +
  stat_summary(aes(group = group), fun = mean, geom = "line",
linewidth = 1.5, na.rm = TRUE) +
  stat_summary(aes(group = group), fun = mean, geom = "point",
size = 2.5, na.rm = TRUE) +
  labs(
    title = "Calcium Levels Over Time by Treatment Group",
    x = "Year",
    y = "Calcium Level",
    color = "Group"
  ) +
  theme_minimal()
```

Usage 2: Formatting the Plots

Date of Usage

April 13, 2026

Prompts Used

“How can I hide the code on a Quarto Document but still have the graphs visible? also how can I widen the graph so it takes up more space on the page?”

Output

Usage 3: Debugging Quarto Errors

Date of Usage

April 14, 2026

Prompts Used

“The following table has numbers out of order. How can I reorganize it?”

```
mutate(
      year = year,
      passengers = readr::parse_number(passengers_raw)
    ) |>
    select(airport, year, passengers)
```

```
}
```

Output

```
  select(airport, '2020', '2021', '2022', '2023', '2024',
'2025') |>
  arrange(airport) |>
  mutate(across(-airport, scales::comma))
```